

```
import pandas as pd
import numpy as np
import string
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity
import nltk
from nltk.corpus import stopwords, wordnet
from nltk.tokenize import word_tokenize
from nltk.stem import WordNetLemmatizer
```

```
nltk.download('punkt')
nltk.download('stopwords')
nltk.download('wordnet')
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
[nltk_data] Downloading package stopwords to /root/nltk_data...
[nltk_data]   Unzipping corpora/stopwords.zip.
[nltk_data] Downloading package wordnet to /root/nltk_data...
True
```

```
data = {
    "Sentence1": [
        "The cat sits on the mat",
        "He is playing football",
        "She likes reading books",
        "The sky is blue",
        "Dogs are loyal animals",

        "A doctor treats patients",
        "The car is very fast",
        "She is cooking food",
        "He bought a new phone",
        "The sun is bright",

        "The boy is happy",
        "She is writing an exam",
        "The teacher explains the lesson",
        "He is running quickly",
        "She loves classical music",

        "I am eating an apple",
        "The train arrived late",
        "It is raining heavily",
        "He opened the window",
        "The book is on the table",

        "She painted a picture",
        "The mobile battery is low",
        "He repaired the bike",
        "The flowers are beautiful",
        "She watched a movie",

        "The computer is expensive",
        "He drinks coffee daily",
```

```

        "The road is crowded",
        "She woke up early",
        "The dog barked loudly"
    ],
    "Sentence2": [
        "The cat sits on the mat",
        "He is playing football",
        "She likes reading books",
        "The sky is blue",
        "Dogs are loyal animals",

        "A physician helps sick people",
        "The vehicle is quick",
        "She prepares meals",
        "He purchased a smartphone",
        "The sunlight is strong",

        "The child feels joyful",
        "She is attempting a test",
        "The instructor teaches the topic",
        "He runs very fast",
        "She enjoys instrumental music",

        "Cars are parked outside",
        "She is cooking dinner",
        "The cat is sleeping",
        "Music makes people happy",
        "He lost his keys",

        "The sky looks cloudy",
        "Charging is required",
        "She bought new shoes",
        "The exam was difficult",
        "They are playing cricket",

        "Prices are increasing",
        "Tea is very hot",
        "The classroom is silent",
        "Birds are flying",
        "Children are laughing"
    ]
}

df = pd.DataFrame(data)
df

```



|    | Sentence1                       | Sentence2                        |
|----|---------------------------------|----------------------------------|
| 0  | The cat sits on the mat         | The cat sits on the mat          |
| 1  | He is playing football          | He is playing football           |
| 2  | She likes reading books         | She likes reading books          |
| 3  | The sky is blue                 | The sky is blue                  |
| 4  | Dogs are loyal animals          | Dogs are loyal animals           |
| 5  | A doctor treats patients        | A physician helps sick people    |
| 6  | The car is very fast            | The vehicle is quick             |
| 7  | She is cooking food             | She prepares meals               |
| 8  | He bought a new phone           | He purchased a smartphone        |
| 9  | The sun is bright               | The sunlight is strong           |
| 10 | The boy is happy                | The child feels joyful           |
| 11 | She is writing an exam          | She is attempting a test         |
| 12 | The teacher explains the lesson | The instructor teaches the topic |
| 13 | He is running quickly           | He runs very fast                |
| 14 | She loves classical music       | She enjoys instrumental music    |
| 15 | I am eating an apple            | Cars are parked outside          |
| 16 | The train arrived late          | She is cooking dinner            |
| 17 | It is raining heavily           | The cat is sleeping              |

```
stop_words = set(stopwords.words('english'))
lemmatizer = WordNetLemmatizer()

def preprocess(text):
    text = text.lower()
    text = text.translate(str.maketrans('', '', string.punctuation))
    tokens = word_tokenize(text)
    tokens = [w for w in tokens if w not in stop_words]
    tokens = [lemmatizer.lemmatize(w) for w in tokens]
    return " ".join(tokens)
```

```
import nltk
nltk.download('punkt_tab')
df["Clean1"] = df["Sentence1"].apply(preprocess)
df["Clean2"] = df["Sentence2"].apply(preprocess)
df[["Clean1", "Clean2"]]
```



```
[nltk_data] Downloading package punkt_tab to /root/nltk_data...  
[nltk_data]   Package punkt_tab is already up-to-date!
```

|    | Clean1                 | Clean2                     |
|----|------------------------|----------------------------|
| 0  | cat sits mat           | cat sits mat               |
| 1  | playing football       | playing football           |
| 2  | like reading book      | like reading book          |
| 3  | sky blue               | sky blue                   |
| 4  | dog loyal animal       | dog loyal animal           |
| 5  | doctor treat patient   | physician help sick people |
| 6  | car fast               | vehicle quick              |
| 7  | cooking food           | prepares meal              |
| 8  | bought new phone       | purchased smartphone       |
| 9  | sun bright             | sunlight strong            |
| 10 | boy happy              | child feel joyful          |
| 11 | writing exam           | attempting test            |
| 12 | teacher explain lesson | instructor teach topic     |

```
vectorizer = TfidfVectorizer()  
tfidf_matrix = vectorizer.fit_transform(  
    df["Clean1"].tolist() + df["Clean2"].tolist()  
)
```

```
cosine_scores = []  
for i in range(len(df)):  
    vec1 = tfidf_matrix[i]  
    vec2 = tfidf_matrix[i + len(df)]  
    score = cosine_similarity(vec1, vec2)[0][0]  
    cosine_scores.append(score)  
df["Cosine_Similarity"] = cosine_scores  
df[["Sentence1", "Sentence2", "Cosine_Similarity"]]
```



|    | Sentence1                       | Sentence2                        | Cosine_Similarity |
|----|---------------------------------|----------------------------------|-------------------|
| 0  | The cat sits on the mat         | The cat sits on the mat          | 1.000000          |
| 1  | He is playing football          | He is playing football           | 1.000000          |
| 2  | She likes reading books         | She likes reading books          | 1.000000          |
| 3  | The sky is blue                 | The sky is blue                  | 1.000000          |
| 4  | Dogs are loyal animals          | Dogs are loyal animals           | 1.000000          |
| 5  | A doctor treats patients        | A physician helps sick people    | 0.000000          |
| 6  | The car is very fast            | The vehicle is quick             | 0.000000          |
| 7  | She is cooking food             | She prepares meals               | 0.000000          |
| 8  | He bought a new phone           | He purchased a smartphone        | 0.000000          |
| 9  | The sun is bright               | The sunlight is strong           | 0.000000          |
| 10 | The boy is happy                | The child feels joyful           | 0.000000          |
| 11 | She is writing an exam          | She is attempting a test         | 0.000000          |
| 12 | The teacher explains the lesson | The instructor teaches the topic | 0.000000          |
| 13 | He is running quickly           | He runs very fast                | 0.000000          |
| 14 | She loves classical music       | She enjoys instrumental music    | 0.262215          |
| 15 | I am eating an apple            | Cars are parked outside          | 0.000000          |
| 16 | The train arrived late          | She is cooking dinner            | 0.000000          |
| 17 | It is raining heavily           | The cat is sleeping              | 0.000000          |
| 18 | He opened the window            | Music makes people happy         | 0.000000          |
| 19 | The book is on the table        | He lost his keys                 | 0.000000          |
| 20 | She painted a picture           | The sky looks cloudy             | 0.000000          |
| 21 | The mobile battery is low       | Charging is required             | 0.000000          |
| 22 | He repaired the bike            | She bought new shoes             | 0.000000          |
| 23 | The flowers are beautiful       | The exam was difficult           | 0.000000          |
| 24 | She watched a movie             | They are playing cricket         | 0.000000          |
| 25 | The computer is expensive       | Prices are increasing            | 0.000000          |
| 26 | He drinks coffee daily          | Tea is very hot                  | 0.000000          |
| 27 | The road is crowded             | The classroom is silent          | 0.000000          |

```
def jaccard_similarity(s1, s2):
    set1 = set(s1.split())
```

```
set2 = set(s2.split())  
return len(set1 & set2) / len(set1 | set2)
```

```
df["Jaccard_Similarity"] = df.apply(  
    lambda row: jaccard_similarity(row["Clean1"], row["Clean2"]),  
    axis=1  
)  
  
df[["Sentence1", "Sentence2", "Jaccard_Similarity"]]
```



|    | Sentence1                       | Sentence2                        | Jaccard_Similarity |
|----|---------------------------------|----------------------------------|--------------------|
| 0  | The cat sits on the mat         | The cat sits on the mat          | 1.0                |
| 1  | He is playing football          | He is playing football           | 1.0                |
| 2  | She likes reading books         | She likes reading books          | 1.0                |
| 3  | The sky is blue                 | The sky is blue                  | 1.0                |
| 4  | Dogs are loyal animals          | Dogs are loyal animals           | 1.0                |
| 5  | A doctor treats patients        | A physician helps sick people    | 0.0                |
| 6  | The car is very fast            | The vehicle is quick             | 0.0                |
| 7  | She is cooking food             | She prepares meals               | 0.0                |
| 8  | He bought a new phone           | He purchased a smartphone        | 0.0                |
| 9  | The sun is bright               | The sunlight is strong           | 0.0                |
| 10 | The boy is happy                | The child feels joyful           | 0.0                |
| 11 | She is writing an exam          | She is attempting a test         | 0.0                |
| 12 | The teacher explains the lesson | The instructor teaches the topic | 0.0                |
| 13 | He is running quickly           | He runs very fast                | 0.0                |
| 14 | She loves classical music       | She enjoys instrumental music    | 0.2                |
| 15 | I am eating an apple            | Cars are parked outside          | 0.0                |
| 16 | The train arrived late          | She is cooking dinner            | 0.0                |
| 17 | It is raining heavily           | The cat is sleeping              | 0.0                |
| 18 | He opened the window            | Music makes people happy         | 0.0                |
| 19 | The book is on the table        | He lost his keys                 | 0.0                |
| 20 | She painted a picture           | The sky looks cloudy             | 0.0                |
| 21 | The mobile battery is low       | Charging is required             | 0.0                |
| 22 | He repaired the bike            | She bought new shoes             | 0.0                |
| 23 | The flowers are beautiful       | The exam was difficult           | 0.0                |
| 24 | She watched a movie             | They are playing cricket         | 0.0                |
| 25 | The computer is expensive       | Prices are increasing            | 0.0                |
| 26 | He drinks coffee daily          | Tea is very hot                  | 0.0                |

```
def wordnet_similarity(s1, s2):
    words1 = s1.split()
```

```
words2 = s2.split()
scores = []

for w1 in words1:
    for w2 in words2:
        syn1 = wordnet.synsets(w1)
        syn2 = wordnet.synsets(w2)
        if syn1 and syn2:
            sim = syn1[0].wup_similarity(syn2[0])
            if sim:
                scores.append(sim)
return np.mean(scores) if scores else 0
```

```
df["WordNet_Similarity"] = df.apply(
    lambda row: wordnet_similarity(row["Clean1"], row["Clean2"]),
    axis=1
)

df[["Sentence1", "Sentence2", "WordNet_Similarity"]]
```



|    | Sentence1                       | Sentence2                        | WordNet_Similarity |
|----|---------------------------------|----------------------------------|--------------------|
| 0  | The cat sits on the mat         | The cat sits on the mat          | 0.480689           |
| 1  | He is playing football          | He is playing football           | 0.804348           |
| 2  | She likes reading books         | She likes reading books          | 0.479012           |
| 3  | The sky is blue                 | The sky is blue                  | 0.566667           |
| 4  | Dogs are loyal animals          | Dogs are loyal animals           | 0.601323           |
| 5  | A doctor treats patients        | A physician helps sick people    | 0.272512           |
| 6  | The car is very fast            | The vehicle is quick             | 0.312745           |
| 7  | She is cooking food             | She prepares meals               | 0.333600           |
| 8  | He bought a new phone           | He purchased a smartphone        | 0.517949           |
| 9  | The sun is bright               | The sunlight is strong           | 0.253788           |
| 10 | The boy is happy                | The child feels joyful           | 0.311111           |
| 11 | She is writing an exam          | She is attempting a test         | 0.235931           |
| 12 | The teacher explains the lesson | The instructor teaches the topic | 0.272141           |
| 13 | He is running quickly           | He runs very fast                | 0.213786           |
| 14 | She loves classical music       | She enjoys instrumental music    | 0.390123           |
| 15 | I am eating an apple            | Cars are parked outside          | 0.263012           |
| 16 | The train arrived late          | She is cooking dinner            | 0.173896           |
| 17 | It is raining heavily           | The cat is sleeping              | 0.144311           |
| 18 | He opened the window            | Music makes people happy         | 0.219774           |
| 19 | The book is on the table        | He lost his keys                 | 0.344300           |
| 20 | She painted a picture           | The sky looks cloudy             | 0.208081           |
| 21 | The mobile battery is low       | Charging is required             | 0.153617           |
| 22 | He repaired the bike            | She bought new shoes             | 0.262567           |
| 23 | The flowers are beautiful       | The exam was difficult           | 0.238492           |
| 24 | She watched a movie             | They are playing cricket         | 0.269048           |
| 25 | The computer is expensive       | Prices are increasing            | 0.200298           |
| 26 | He drinks coffee daily          | Tea is very hot                  | 0.314298           |

df

|   | Sentence1                      | Sentence2                           | Clean1                     | Clean2                           | Cosine_Similarity | Jaccard_ |
|---|--------------------------------|-------------------------------------|----------------------------|----------------------------------|-------------------|----------|
| 0 | The cat sits<br>on the mat     | The cat sits<br>on the mat          | cat sits<br>mat            | cat sits mat                     | 1.000000          |          |
| 1 | He is<br>playing<br>football   | He is<br>playing<br>football        | playing<br>football        | playing<br>football              | 1.000000          |          |
| 2 | She likes<br>reading<br>books  | She likes<br>reading<br>books       | like<br>reading<br>book    | like reading<br>book             | 1.000000          |          |
| 3 | The sky is<br>blue             | The sky is<br>blue                  | sky blue                   | sky blue                         | 1.000000          |          |
| 4 | Dogs are<br>loyal<br>animals   | Dogs are<br>loyal<br>animals        | dog loyal<br>animal        | dog loyal<br>animal              | 1.000000          |          |
| 5 | A doctor<br>treats<br>patients | A physician<br>helps sick<br>people | doctor<br>treat<br>patient | physician<br>help sick<br>people | 0.000000          |          |
|   | ...                            | ...                                 |                            | ...                              |                   |          |